

Run 01 | TEA

Global (including India)

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 154,420 (11.50%) **Case time** 50m 07s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9913). Highest F1: Random Forest (0.8657). The all-negative accuracy baseline is 88.50% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9913	0.9288	96.84	84.69	88.53	0.8657	2m 51s
XGBoost	0.9913	0.9292	96.79	84.20	88.75	0.8641	32s
CatBoost	0.9910	0.9267	96.75	83.80	88.90	0.8627	2m 28s
H2O GBM	0.9897	0.9129	96.44	80.48	91.10	0.8546	38s
Lasso	0.9886	0.8996	96.32	79.97	90.67	0.8498	11m 39s
Elastic Net	0.9886	0.8996	96.31	79.95	90.67	0.8497	14m 05s
Logistic Regression	0.9885	0.8996	96.31	79.95	90.69	0.8498	2m 01s
Ridge	0.9884	0.8984	96.29	79.89	90.56	0.8489	10m 46s
Decision Tree	0.9853	0.8396	96.24	80.74	88.39	0.8439	13s
Naive Bayes	0.9210	0.4838	89.02	51.37	84.94	0.6402	4m 21s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	81,565.98	55.23
2	Established business owner	28,976.11	19.62
3	Self-efficacy	4,587.96	3.11
4	Knowing entrepreneurs	4,010.95	2.72
5	Size of government	3,813.49	2.58
6	Legal system and property rights	2,994.61	2.03
7	Freedom to trade internationally	2,367.62	1.60
8	Regulation	2,270.16	1.54

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Size of government	3,813.49	2.58
2	Legal system and property rights	2,994.61	2.03
3	Freedom to trade internationally	2,367.62	1.60

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 01_obj3_global_iv_TEA. Scores and importance are taken directly from saved CSV outputs.